

# YASH PATEL

DeLand, FL, 32723 (*Open to relocation*)

386-986-8201 | [ypatel2@stetson.edu](mailto:ypatel2@stetson.edu) | [yashbpatel.vercel.app](https://yashbpatel.vercel.app) | [github.com/YashPatel04](https://github.com/YashPatel04) | [linkedin.com/in/yashpatel2004](https://linkedin.com/in/yashpatel2004)

## EDUCATION

Stetson University, DeLand, FL

*Expected Graduation: Dec 2026*

*Bachelor of Science, Computer Science (Dean's List Student)*

GPA: 3.77

**Relevant Coursework:** Calculus I, II & III, Software Development I & II, Computer Networks, Object-Oriented Design, Web Development, Database Systems, Algorithm Analysis, Machine Learning, Operating Systems, Artificial Intelligence

## TECHNICAL SKILLS

**Programming Languages:** Python, JavaScript, TypeScript, Java, C, C++, HTML, PHP, CSS, TailwindCSS

**Frameworks & Libs:** React, Redux, Next.js, Node.js, Express, Django, OpenCV, LangChain, SciKit, TensorFlow, PyTorch

**Cloud & DevOps:** AWS (EC2, S3, RDS), GCP, Docker, Kubernetes, SQL, MongoDB, Linux, NGINX, Redis, PostgreSQL

**Tools & Practices:** NPM, PM2, Git, REST APIs, SDLC, Agile, Postman, Unit Testing, CI/CD, RAG, Ollama, n8n

**Certifications:** Python ([Link](#)), C++ ([Link](#)), Data Structures and Algorithms (Java) ([Link](#))

## EXPERIENCE

Stetson University

August 2025 – December 2025

*Software Engineer Intern (Capstone Proj.)*

*DeLand, FL*

- Collaborated with a **8-member** cross-disciplinary team to design and deploy a **cloud-native virtualized campus** network using AWS EC2 and Docker, simulating **50+ concurrent cyberattack scenarios** to test network resilience.
- Deployed Suricata IDS/IPS with **45K+ active detection rules** and integrated Cowrie **honeypots** across distributed systems, improving threat visibility and enabling automated correlation of intrusion events.
- Fine-tuned **RAG-based LLM** agents with vector stores (ChromaDB) and web scraping, **cutting false positives by 40%** and enhancing reliability in AI-driven incident detection.

SignStream - Digital Signage Solution

Dec 2024 – Present

*Founder & Developer*

[signstreamus.com](https://signstreamus.com)

- Independently designed and deployed a cloud-based **digital signage SaaS platform** serving **5 retail clients**, enabling remote content scheduling and achieving **99% uptime** with real-time dynamic updates.
- Engineered secure and fault-tolerant media streaming with Socket.io and edge caching on Single Board Computers (SBC), deploying backend services on AWS EC2 with firewall rules, SSL encryption, and rate-limited APIs, serving **500 + daily ad impressions**.

AASOA - Store Owners Association

May 2024 – July 2024

*Full Stack Intern (Contract)*

*Remote(India)*

- Developed and maintained the company's **WordPress portal**, building custom **PHP-based webpages and forms** that processed over **2,000+ user registrations**.

## RESEARCH

Prompt Optimization for LLMs via Squirrel Search Algorithm

Oct 2025 – Jan 2026

*Lead Author & Researcher*

*Accepted at ACM Southeast Conference (ACMSE'26, Troy, AL) | [Paper](#)*

- Developed a **novel gradient-free SSA implementation** for automated prompt optimization, achieving **57% improvements** in sentiment classification across multiple LLMs (3.8B–20B parameters) with convergence within **10–30 iterations**.

FACETS: Multi-Dimensional LLM Benchmarking Framework

Jan 2026 – Present

*Lead Author & Researcher*

*Stetson University Senior Research, Advisor: Dr. Basar Koc*

- Designed **FACETS**, a modular benchmarking framework evaluating **13 LLMs** across **6 software engineering tasks** and **9 quality dimensions**, revealing a **24.77-point performance spread** (64.56%–89.33%) and a persistent gap between syntactic fluency and semantic understanding in code generation.

Automated Attendance via Face Recognition

Dec 2025 – Present

*Author & Researcher*

*Stetson University*

- Researched and combined multiple datasets using Python to train an **image segmentation model**; fine-tuned a **DETR (Detection Transformer)** achieving **90% accuracy**, enabling real-time face identification for automated classroom attendance workflows.

## PROJECTS

Domain-Specific AI Chatbot | *LangFlow, Ollama, ChromaDB, Google Gemini, Docker* | [GitHub Repo](#)

Nov 2025

- Built a **RAG-based chatbot** for car dealership customers using Ollama embeddings and ChromaDB semantic search; integrated Google Gemini LLM with conversation memory to deliver **contextually-accurate responses** grounded in dealership documents.

CliquePay | *Django, React, AWS, MySQL* | [GitHub Repo](#)

Jan 2025 – May 2025

- Led a team of **3 developers** to architect and deploy a **React + Django** full-stack finance app; implemented scalable **REST APIs**, integrated AWS Cognito authentication, optimized SQL queries via Django ORM to **reduce database latency by 40%**, and built **real-time group chats** using WebSockets, deployed on AWS with Docker and coordinated via Agile sprints for on-schedule delivery.